

JUNIOR CANADIAN CHEMISTRY OLYMPIAD

British Columbia Chemistry Olympiad

BCCO 2019

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

考试期间允许使用元素周期表/化学数据表以及非编程计算器。

Write your answers on the multiple choice answer sheet provided. Use a pencil.

将答案用铅笔填涂在答题卡上。

Students are expected to complete the examination in 1.5 hours.

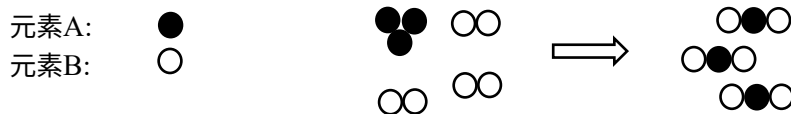
考试时间为90分钟。

- 1) Which of the following chemical equations is properly balanced?
以下哪个化学方程式已经配平?
 - a) $\text{CH}_3\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 - b) $2\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{Na}_2\text{HPO}_4 + \text{H}_2\text{O}$
 - c) $\text{Ba}(\text{NO}_3)_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{NaNO}_3$
 - d) $\text{C}_6\text{H}_8 + 8\text{O}_2 \rightarrow 6\text{CO}_2 + 4\text{H}_2\text{O}$
 - e) $3\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{Na}_3\text{PO}_4 + 2\text{H}_2\text{O}$
- 2) Camphor ($\text{C}_{10}\text{H}_{16}\text{O}$) is a terpenoid with a strong aroma found in camphor trees leaves. If an average of 45.6 mg of camphor can be extracted from leaf of camphor tree, how many moles of camphor can be extracted from 100 camphor tree leaves?
樟脑($\text{C}_{10}\text{H}_{16}\text{O}$)是一种存在于樟树叶子中具有强烈香气的萜类化合物。一片樟树叶子中可提取平均45.6 mg的樟脑,那么从100片樟树叶子中可提取多少摩尔的樟脑?
 - a) 4.56×10^{-2}
 - b) 3.00×10^{-2}
 - c) 3.00×10^{-3}
 - d) 4.56×10^1
 - e) 3.00×10^{-1}
- 3) Calculate the mass of anhydrous sodium bisulfate (NaHSO_4) required to prepare a 0.10 M bisulfate solution in water using a 100 mL volumetric flask?
在100 mL容量瓶中制备0.10M (0.10mol/L) 硫酸氢钠溶液需要多少质量的无水硫酸氢钠 (NaHSO_4) ?
 - a) 1.19 g
 - b) 1.04 g
 - c) 11.9 g
 - d) 1.20 g
 - e) 0.12 g
- 4) In order to plot a two component phase diagram 0.527 g of naphthalene (128.174 g/mol) and 1.264 g of diphenylamine (169.227 g/mol) are mixed and melted together. What is the mole fraction of naphthalene in the mixture?
为了绘制二组分相图,将0.527g萘 (128.174g / mol) 和1.264g二苯胺 (169.227g / mol) 混合并熔融在一起。混合物中萘的摩尔分数是多少?
 - a) 0.550
 - b) 0.645
 - c) 0.355
 - d) 0.450
 - e) 0.569
- 5) The first law of Thermodynamics states that
热力学第一定律指出
 - a) Entropy is zero for a perfect crystal of a pure substance at zero Kelvin.
在绝对零度 (零开尔文) 时,任何纯物质的完美晶体的熵值为零
 - b) The entropy of a system increases for any spontaneous process.
对于任何自发过程,系统的熵都会增加
 - c) An object maintains its state of motion unless acted upon by an unbalanced force.
除非受到不平衡的作用力,否则物体会保持其运动状态
 - d) Heat flows spontaneously from a hot body to a cold body.
热量自发地从高温物体传递到低温物体
 - e) Energy is conserved
能量是守恒的

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- 6) What is the balanced form of the chemical reaction depicted below?

下图表达的化学反应平衡方程式是什么？



- a) $A_3 + 3B_2 \rightarrow 3AB_2$
b) $3A + 3B_2 \rightarrow 3AB_2$
c) $3A + 6B \rightarrow 3AB_2$
d) $3A + 6B \rightarrow A_3B_6$
e) $A_3 + B_2 \rightarrow AB_2$
- 7) The density of ethanol (C_2H_5OH) at $20^\circ C$ is 0.7893 g/cm^3 . What is the volume of 1 mole of ethanol at $20^\circ C$?

已知 $20^\circ C$ 时，乙醇 (C_2H_5OH) 的密度为 0.7893 g/cm^3 ，在 $20^\circ C$ ，1摩尔乙醇的体积是多少？

- a) 46.1 mL b) 58.4 mL c) 36.4 mL d) 0.584 L e) 46.1 m^3
- 8) Which of the following accurately represents a trend in electronegativity?
下列关于电负性的排序哪项是正确的？
- a) $Ne > F > O$ b) $F > O > N$ c) $Br > Cl > F$ d) $Cs > Rb > Na$ e) $Al > P > S$
- 9) Which of the following accurately represents a trend in the first ionization energies?
下列关于第一电离能的排序哪项是正确的？

- a) $Ne > F > C$ b) $Br > Cl > F$ c) $Cs > Rb > Na$ d) $Ar > Ne > He$ e) $Al > P > S$

- 10) The second law of Thermodynamics states that

热力学第二定律指出

- a) Entropy is zero for a perfect crystal of a pure substance at zero Kelvin.
在绝对零度（零开尔文）时，任何纯物质的完美晶体的熵值为零。
- b) The entropy of a system increases for any spontaneous process.
对于任何自发过程，系统的熵都会增加。
- c) An object maintains its state of motion unless acted upon by an unbalanced force.
除非受到不平衡的作用力，否则物体会保持其运动状态。
- d) The entropy of the universe is continually increasing.
宇宙中的熵在不断增加。
- e) Energy is conserved.
能量是守恒的。

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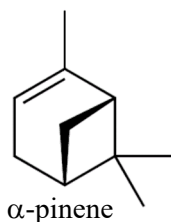
- 11) To study weather phenomena sensors carried by balloons are used. What is the mass of helium that is needed to fill a 65 m^3 balloon to a pressure of 1 atm at 25°C ? Assume ideal gas behavior.

研究天气现象时会使用一种连有传感器的气球。在压力为1 atm，温度为 25°C 时，需要多少氦气才能将 65 m^3 的气球充满？假设该气体为理想气体。

a) 2.66 kg b) 0.105 g c) 11.61 g d) 10.63 kg e) 126.8 kg

- 12) α -Pinene is a terpenoid with a strong aroma found in pine resin.

a-蒎烯是一种存在于松树树脂中具有强烈香气的萜烯类化合物。



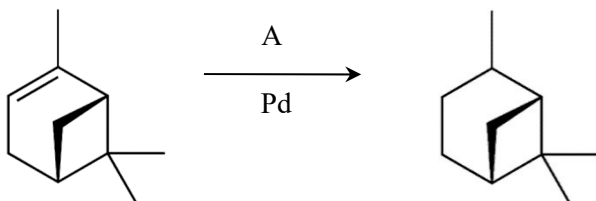
Which of the following molecular formula is the correct one for α -pinene?

以下哪个是a-蒎烯的正确分子式？

a) C_9H_{11} b) $\text{C}_{10}\text{H}_{14}$ c) $\text{C}_{10}\text{H}_{16}$ d) C_9H_{14} e) C_9H_{16}

- 13) α -Pinene is a reactant in the following reaction:

a-蒎烯是下列反应的反应物：



Which of the following compounds could be used as reagent A?

下列哪种化合物可以用作试剂A？

a) HCl b) H_2 c) O_3 d) H_2O e) NaHCO_3

- 14) A solution of tetrachloroethane ($\text{C}_2\text{H}_2\text{Cl}_4$) in cyclohexanone ($\text{C}_6\text{H}_{10}\text{O}$) is prepared with 29.3 mL of tetrachloroethane in 86.5 mL of cyclohexanone. What is the mole fraction of tetrachloroethane in the solution? Assume densities of 1.59 g/cm^3 for tetrachloroethane and 0.948 g/cm^3 for cyclohexanone.

用29.3 mL四氯乙烷 ($\text{C}_2\text{H}_2\text{Cl}_4$) 和86.5 mL环己酮 ($\text{C}_6\text{H}_{10}\text{O}$) 配制环己酮和四氯乙烷的混合溶液。求混合溶液中四氯乙烷的摩尔分数是多少？假设四氯乙烷的密度为 1.59 g/cm^3 ，环己酮的密度为 0.948 g/cm^3 。

a) 0.249 b) 0.165 c) 0.751 d) 0.835 e) 0.181

- 15) A catalyst cannot alter

催化剂不能改变

a) the rate of a reaction.
b) the activation energy of a reaction
c) the equilibrium constant of a reaction
d) the half-life of a reaction
e) the rate of a reaction

a) 反应速率
b) 反应活化能
c) 反应的平衡常数
d) 反应的半衰期
e) 反应的速率常数

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- 16) ^{14}C is eliminated from the body of a dead mammoth by a first order process that has a half-life of 5750 years. What percentage of the initial ^{14}C would remain in the mammoth body after 11 500 years?

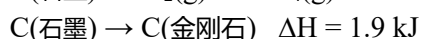
在一级动力学中，猛犸象尸体中的 ^{14}C 的半衰期是5750年。11500年后，最初的 ^{14}C 会有多少残留在猛犸象体内？

a) 0% b) 25% c) 50% d) 75% e) 2%

- 17) Consider the following reaction: 考虑反应: $\text{CH}_4(\text{g}) \rightarrow \text{C}(\text{金刚石}) + 2\text{H}_2(\text{g})$.

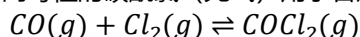
Use the information provided below to calculate the enthalpy of reaction when 1 mole of diamond is formed.

利用下面提供的信息计算生成1摩尔金刚石时的反应焓。



a) -76.7 kJ b) 72.9 kJ c) 76.7 kJ d) -72.9 kJ e) 74.8 kJ

- 18) The highly toxic carbonyl chloride (phosgene) is used to synthesize many organic compounds: 高毒性的碳酰氯（光气）用于合成许多有机化合物：



The equilibrium constant K_p for the gas-phase synthesis of phosgene is 7.70 at 75 °C.

What is the equilibrium pressure of COCl_2 if the equilibrium pressures of CO and Cl_2 are both 2 atm?

在75°C下气相合成光气的平衡常数 K_p 为7.70。如果 CO 和 Cl_2 的平衡分压均为2 atm， COCl_2 的平衡压力是多少？

a) 1.90 atm b) 15.4 atm c) 30.8 atm d) 3.90 atm e) 237 atm

- 19) The rate law for the above reaction is as follows: $v = k[\text{CO}][\text{Cl}_2]^x$

Use the data table below to find the value of x.

实验	初始浓度 (M)		初始速率 (Ms^{-1})
	[CO]	[Cl ₂]	
1	0.12	0.20	0.121
2	0.24	0.40	0.684

上述反应的速率方程: $v = k[\text{CO}][\text{Cl}_2]^x$

使用上表中的数据求出x值

a) 1.0 b) 0.5 c) 0 d) 2.0 e) 1.5

- 20) When 7.41g of solid $\text{Ca}(\text{OH})_2$ (Molar mass = 74.09g/mole, $K_{sp} = 5.02 \times 10^{-6}$) is dissolved into 1.00L of water (assume there is no change in volume), the pH of the solution is closest to,

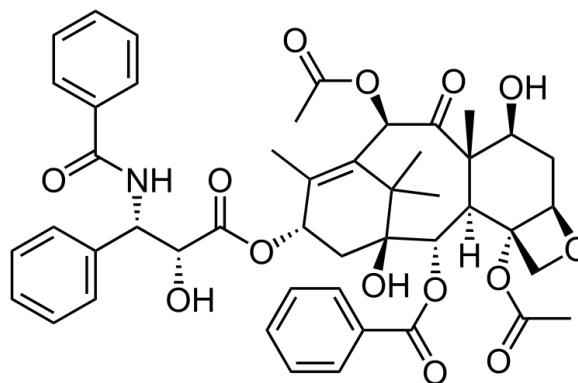
当将7.41g固体 $\text{Ca}(\text{OH})_2$ （摩尔质量= 74.09g / mol, $K_{sp} = 5.02 \times 10^{-6}$ ）溶于1.00L水中（假设体积不变）时，溶液的pH值最接近？

a) 13.00 b) 12.67 c) 12.33 d) 12.00 e) 11.67

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- 21) Taxol is a natural product isolated from the Pacific Yew tree in the north western part of the province of British Columbia and has been used as a drug in the treatment of a few types of cancers. The structure of Taxol molecule is shown on the right. Based on the structure, a Taxol molecule contains the following functional groups,

紫杉醇是从英国哥伦比亚省西北部的太平洋紫杉中提取出来的天然产物，已被用作治疗多种癌症的药物。紫杉醇分子的结构如右图所示。基于该结构，紫杉醇分子包含以下哪些官能团？



- a) Alcohol, Aldehyde, Amide, Ester, Ketone
- b) Alcohol, Amide, Ester, Ether, Ketone
- c) Alcohol, Aldehyde, Amine, Ester, Ether
- d) Alcohol, Amine, Ketone, Ester, Ether
- e) Alcohol, Aldehyde, Amide, Ester

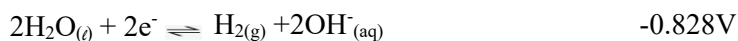
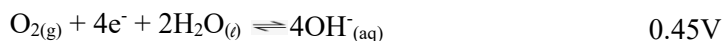
- a) 羟基, 醛基, 酰胺键, 酯基, 酮羰基
- b) 羟基, 酰胺键, 酯基, 醚键, 酮羰基
- c) 羟基, 醛基, 氨基, 酯基, 醚键
- d) 羟基, 氨基, 酮羰基, 酯基, 醚键
- e) 羟基, 醛基, 酰胺键, 酯基

- 22) Household bleach can be produced by the electrolysis of sea water. Given the standard reduction potential as follows,

家用漂白剂可以通过电解海水来制得。下列所示是标准还原电势

What is produced at the negative electrode during the electrolysis of sea water,

电解海水过程中在负极产生的物质为？



- a) $\text{Cl}_{2(\text{g})}$
- b) $\text{O}_{2(\text{g})}$
- c) $\text{Na}_{(\text{s})}$
- d) $\text{H}_{2(\text{g})}$
- e) $\text{NaClO}_{(\text{s})}$

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- 23) Which of the following is ranked correctly based on the increasing melting point under the standard atmospheric pressure,

在标准大气压下，以下哪一项是按照熔点的升高进行排序的？

- a) $F_2 < Ga < CH_3CH_2OH < NaCl < SiO_2$
- b) $F_2 < CH_3CH_2OH < Ga < SiO_2 < NaCl$
- c) $F_2 < CH_3CH_2OH < NaCl < SiO_2 < Ga$
- d) $F_2 < Ga < CH_3CH_2OH < SiO_2 < NaCl$
- e) $F_2 < CH_3CH_2OH < Ga < NaCl < SiO_2$

- 24) Based on the VSEPR theory, in how many of the following molecular geometries are all atoms lay on the same plane,

根据VSEPR（价层电子对互斥）理论，在下面的分子几何构型中，有几个分子几何构型能够满足所有原子都位于同一平面上？

Linear 直线型

Trigonal planar 平面三角形

Bent V 字型

T-shape T 字型

Square planar 平面正方形

- a) 1 b) 2 c) 3 d) 4 e) 5

- 25) What will be the pH of a solution prepared by mixing 100. mL of 0.50M acetic acid with 50. mL of 0.50M NaOH. The K_a of acetic acid is 1.8×10^{-5} ,

将100 mL的0.50 M (0.50mol/L) 乙酸与50 mL的0.50 M NaOH混合制备的溶液的pH值是多少？

(乙酸的 K_a 为 1.8×10^{-5})

- a) 4.74 b) 9.26 c) 2.76 d) 8.99 e) 5.01

1	H	1.008	Hydrogen
3	Li	6.94	Lithium
11	Na	22.98976928	Sodium
19	K	39.0983	Potassium
37	Rb	85.4678	Rubidium
55	Cs	132.90545196	Caesium
87	Fr	223	Francium

Data Sheet –Periodic Table and Constants

原子序数

Atomic Number

→ 1

元素符号

Symbol

→ H

1.008

← Atomic Mass

相对原子质量

Hydrogen

← Name

元素名称

4	Be	9.0121831	Beryllium
12	Mg	24.305	Magnesium
20	Ca	40.078	Calcium
38	Sr	87.62	Strontium
56	Ba	137.327	Barium
88	Ra	226	Radium
21	Sc	44.955908	Scandium
22	Ti	47.867	Titanium
23	V	50.9415	Vanadium
24	Cr	51.9961	Chromium
25	Mn	54.938044	Manganese
26	Fe	55.845	Iron
27	Co	58.933194	Cobalt
28	Ni	58.6934	Nickel
29	Cu	63.546	Copper
30	Zn	65.38	Zinc
31	Ga	69.723	Gallium
32	Ge	72.630	Germanium
33	As	74.921595	Arsenic
34	Se	78.971	Selenium
35	Br	79.904	Bromine
36	Kr	83.798	Krypton
39	Y	88.90584	Yttrium
40	Zr	91.224	Zirconium
41	Nb	92.90637	Niobium
42	Mo	95.95	Molybdenum
43	Tc	98	Technetium
44	Ru	101.07	Ruthenium
45	Rh	102.90550	Rhodium
46	Pd	106.42	Palladium
47	Ag	107.8682	Silver
48	Cd	112.414	Cadmium
49	In	114.818	Indium
50	Sn	118.710	Tin
51	Sb	121.760	Antimony
52	Te	127.60	Tellurium
53	I	126.90447	Iodine
54	Xe	131.293	Xenon
57	71		
57	71		
58	Ce	140.116	Cerium
59	Pr	140.90766	Praseodymium
60	Nd	144.242	Neodymium
61	Pm	145	Promethium
62	Sm	150.36	Samarium
63	Eu	151.964	Europium
64	Gd	157.25	Gadolinium
65	Tb	158.92535	Terbium
66	Dy	162.500	Dysprosium
67	Ho	164.93033	Holmium
68	Er	167.259	Erbium
69	Tm	168.93422	Thulium
70	Yb	173.054	Ytterbium
71	Lu	174.9668	Lutetium
89	Ac	227	Actinium
90	Th	232.0377	Thorium
91	Pa	231.03588	Protactinium
92	U	238.02891	Uranium
93	Np	237	Neptunium
94	Pu	244	Plutonium
95	Am	243	Americium
96	Cm	247	Curium
97	Bk	247	Berkelium
98	Cf	251	Californium
99	Es	252	Einsteinium
100	Fm	257	Fermium
101	Md	258	Mendelevium
102	No	259	Nobelium
103	Lr	266	Lawrencium
116	Lv	293	Livermorium
117	Uus	294	Ununseptium
118	Uuo	294	Ununoctium

Lanthanide Series
镧系

Actinide Series
锕系

- 原子质量单位
- 阿伏伽德罗常量
- 电子电荷量
- 法拉第常数
- 气体常数/理想气体常数
- 一个电子的质量
- 1.66054×10⁻²⁷ kg
- 6.022×10²³
- 1.60218×10⁻¹⁹ C
- 96485 C mol⁻¹
- 8.3145 J K⁻¹ mol⁻¹
- 9.1094×10⁻³¹ kg

- 光速
- 普朗克常数
- 2.99792458×10⁸ m/s
- 6.62637×10⁻³⁴ Js

- 1 atm = 101325 Pa
- 1 bar = 100000 Pa
- 0 °C = 273.15 K